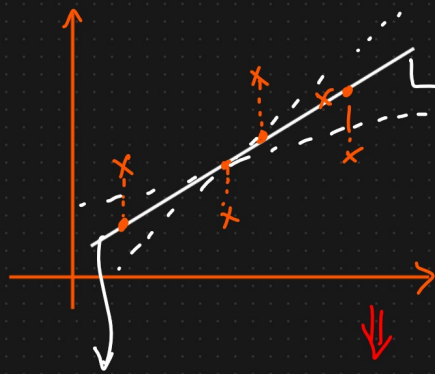
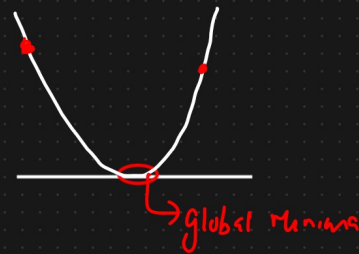


Linear Regression Using OLS {Ordinary Least Square}



optimization
↓
loss function



$h_0(x) = \beta_0 + \beta_1 x$
 $\hookrightarrow \hat{y}$

$\rightarrow$ OLS $\rightarrow$ Apply Formula and Calculate
 β_0 & β_1
 $=$ $=$

Ordinary Least Square

$$S(\beta_0, \beta_1) = \frac{1}{n} \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i)^2$$

↓

Find β_0 & β_1
 $=$ $=$

Error
↑ =

MSR



$$y = x^2$$

$$\frac{\partial y}{\partial x} = 2(x)^{2-1} \frac{\partial (x)}{\partial x} = 2x$$

$$\begin{aligned} \frac{\partial S}{\partial \beta_0}(\beta_0, \beta_1) &= \frac{2}{n} \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i) (0 - 1 - 0) \\ &= -\frac{2}{n} \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i) = 0 \rightarrow \textcircled{1} \checkmark \end{aligned}$$

$$\begin{aligned} \frac{\partial S}{\partial \beta_1}(\beta_0, \beta_1) &= \frac{2}{n} \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i) (-x_i) = 0 \rightarrow \textcircled{2} \\ &= -\frac{2}{n} \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i) (x_i) = 0 \rightarrow \textcircled{2} \checkmark \end{aligned}$$

$$\varepsilon_q \rightarrow \textcircled{1}$$

$$-\frac{2}{n} \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i) = 0$$

$\Downarrow$

$$-\sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i) = 0$$

$$-\sum_{i=1}^n y_i + \overset{\text{for } n \text{ sample}}{n} \beta_0 + \beta_1 \sum_{i=1}^n x_i = 0$$

$$n\beta_0 + \beta_1 \sum_{i=1}^n x_i = \sum_{i=1}^n y_i$$

$$\beta_0 = \frac{\sum_{i=1}^n y_i}{n} - \beta_1 \frac{\sum_{i=1}^n x_i}{n}$$

Intercept

$$\beta_0 = \bar{y} - \beta_1 \bar{x}$$

$$\boxed{\varepsilon_q, 2 \div}$$

$$-\frac{2}{n} \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i)(x_i) = 0$$

$\Downarrow$

$$\sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i)(x_i) = 0$$

$\Downarrow$

$$\sum_{i=1}^n x_i y_i - \beta_0 \sum_{i=1}^n x_i - \beta_1 \sum_{i=1}^n x_i^2 = 0$$

$$\sum_{i=1}^n (x_i y_i - \beta_0 x_i - \beta_1 x_i^2) = 0$$

↓

$$\therefore \text{Replace } \beta_0 = \bar{y} - \beta_1 \bar{x}$$

$$\sum_{i=1}^n (x_i y_i - (\bar{y} - \beta_1 \bar{x}) x_i - \beta_1 x_i^2) = 0$$

$$\sum_{i=1}^n (x_i y_i - x_i \bar{y} + \beta_1 \bar{x} x_i - \beta_1 x_i^2) = 0$$

$$\sum_{i=1}^n (y_i - \bar{y} + \beta_1 \bar{x} - \beta_1 x_i) x_i = 0$$

$$\sum_{i=1}^n [(y_i - \bar{y}) + \beta_1 (\bar{x} - x_i)] = 0$$

$$\sum_{i=1}^n (y_i - \bar{y}) + \sum_{i=1}^n \beta_1 (\bar{x} - x_i) = 0$$

$$\sum_{i=1}^n \beta_1 (\bar{x} - x_i) = - \sum_{i=1}^n (y_i - \bar{y})$$

$$\beta_1 = \frac{- \sum_{i=1}^n (y_i - \bar{y})}{\sum_{i=1}^n (\bar{x} - x_i)}$$

$$\beta_1 = \frac{\sum_{i=1}^n (y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})}$$

Coefficient $\Rightarrow$

$$\beta_1 = \frac{\sum_{i=1}^n (y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})}$$

Intercept
↓

Ols

↑
coefficient

$$\boxed{\beta_0 = \bar{y} - \beta_1 \bar{x}}$$

x	y	$(y_i - \bar{y}) \div (x_i - \bar{x})$	β_1	$\beta_0 = (\bar{y} - \beta_1 \bar{x})$
-	-	-	-	-
-	-	-	-	-
-	-	-	-	-
-	-	-	-	-
$\bar{x}$	$\bar{y}$			

Ols $\approx$ Linear Regression (sklearn)

Ols $\approx$ Linear Regression